VTE Assessment and Prophylaxis in pregnant and postnatal women (GL891)

Approval

Approval Group	Job Title, Chair of Committee	Date
Maternity & Children's Services Clinical Governance Committee	Chair, Maternity Clinical Governance Committee	8th March 2024 & 14/6/24 & 11/10/24
Drugs and Therapeutic Committee	Sakeb Hussain Chief Pharmacist	3rd April 2024

Reviewing Pharmacist (only if the document includes medications)	Date
N Manan – Lead Women, Children & MI Pharmacist	March 2024

Change History

Version	Date	Author, job title	Reason
10.0	June 2023	Stephen Booth – Consultant Haematologist Anand Pankhania – Senior Lead Pharmacist Clinical Services	Switch from tinzaparin to enoxaparin for VTE, extension of enoxaparin to Crcl < 30 ml / min
10.1	March 2024	Stephen Booth – consultant haematologist Wanqing Fan – anticoagulant pharmacist	Updated postnatal thromboprophylaxis risk category and duration. Updated prophylactic enoxaparin dose recommendation for patients ≥ 170kg with Crcl <30ml/min. VTE & Covid criteria amended pg 5
10.2	May 2024	B Chohan, O&G Consultant	Live change to remove reference to Obs advice email pg 3
10.3	July 2024	M Fayyaz, Consultant O&G C Bell, Compliance Lead W Fan – anticoagulant pharmacist	Live change to clarify process for antenatal prophylaxis

Overview: *Thromboembolism remains the leading cause of direct death in the recent MBRRACE reports (MBRRACE 2017, 2018, 2019, 2023). The most important risk factors are* BMI, age above 35 and caesarean delivery

Maternal mortality in UK due to VTE in pregnancy and up to 6 weeks after the end of pregnancy is 1.6 in 100,000 maternities compared to 1.39 in the 2014-2016 MBRRACE report, and 1.01 in 2011-2013 MBRRACE report.

The relative risk of VTE in pregnancy is increased 4-6 fold, with a further increase in risk postpartum, however the absolute incidence of VTE in pregnancy and the puerperium is low, with an overall incidence of 1-3 per 1000.

Many fatal VTE events occur in the first trimester so prophylaxis, if required, should begin in early pregnancy. NICE estimates low molecular weight Heparin (LMWH) reduces VTE risk in medical and surgical patients by 60 and 70% respectively. https://www.npeu.ox.ac.uk/assets/downloads/mbrrace-uk/reports/maternal-report-2023/MBRRACE-UK_Maternal_Compiled_Report_2023.pdf

Risk Assessment

All women will be assessed using the maternity VTE risk assessment form for the risk of VTE at the following times:

- At booking
- On each admission
- With the development of other inter current problems
- Post-delivery (before transfer home/to the ward)

This will be documented on the appropriate proforma which is available on EPR as maternity VTE risk assessment tool (community staff will have access to EPR VTE assessment tool and this will be completed at booking).

Antenatal Prophylaxis

Prophylaxis with antenatal LMWH Enoxaparin is required in the following situations:

1. Women with any previous VTE (except a single event related to major surgery) are considered **high risk** and require antenatal prophylaxis with LMWH Enoxaparin (Inhixa) throughout pregnancy and for 6 weeks post-partum. This treatment needs to be started as early in pregnancy as possible. Patients already on long term oral anticoagulation should be switched immediately to therapeutic enoxaparin if pregnant. GP's will prescribe a 2 week course of LMWH for high risk women (those with prior VTE or on oral anticoagulants) when the woman advises them they have had a positive pregnancy test to "bridge" the period between then and the booking appointment. A pathway is then in place for these women to be referred for an urgent appointment with DAU to obtain the ongoing supply of LMWH. These patients should be referred to Miss Fayyaz's or Miss Gajree's antenatal clinic.

Woman with previous VTE OR on oral anticoagulants has positive pregnancy test

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Woman to contact GP who will provide prescription for 2 week bridging course of LMWH, GP to advise the woman to book with the midwife within 2 weeks

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Woman contacts maternity either by phone or the on-line booking system and needs to identify on the on-line booking form or on the telephone they have had a previous VTE or on oral anticoagulants

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Back office to arrange booking appointment within 2 weeks (if unable to find a booking slot this must be escalated to the community coordinator **urgently and DAU appointment to be booked.**

↓

Community midwife to complete booking and refer woman to consultant lead care (Miss Fayyaz or Miss Gajree)

2. Antenatal prophylaxis should be considered in those patients deemed to be at **intermediate risk**. Please refer to Friday am Haem Obs clinic if required. This includes patients with the following risk factors:

- Hospital admission
- Single previous VTE related to major surgery
- **High-risk thrombophilia + no VTE**
  - Anti-thrombin deficiency
  - Protein C deficiency
  - Protein S deficiency
- More than 1 thrombophilia
- **Medical co-morbidities**
  - Cancer
  - Heart failure
  - Active SLE
  - Inflammatory bowel disease
  - Nephrotic syndrome
  - Type 1 Diabetes with nephropathy
  - Sickle cell disease
  - Current IVDU
- Any surgical procedure during pregnancy
- OHSS (first trimester only)

3. Patients with 4 or more cumulative risk factors (see below) require thromboprophylaxis from the first trimester. This should be arranged through DAU. These patients can be referred to Miss Fayyaz's or Miss Gajree's antenatal clinic if required.

4. Patients with 3 cumulative risk factors (see below) require thromboprophylaxis from 28 weeks. These women can be referred to any consultant antenatal clinic.

5. Patients with fewer than 3 cumulative risk factors (see below) are at **lower risk** and require mobilisation and avoidance of dehydration only.

6. Patients with known anti-thrombin deficiency or APS may need additional 50% or 75% of prophylactic dose. These patients need to be managed in collaboration with the haematologists.

The cumulative risk factors are:

- BMI >30kg/m^2
- BMI >40kg/ m^2 (= 2 risk factors)
- Age >35
- Parity >=3
- Smoker
- Gross Varicose veins
- Current pre-eclampsia
- Immobility (paraplegia, PGP)
- Family history of unprovoked or oestrogen-provoked VTE in 1st degree relative
- Multiple pregnancy
- IVF/ART
- Low-risk thrombophilia
- **Transient risk factors**
    - Dehydration/Hyperemesis (if admitted)
    - Current systemic infection
    - Long-distance travel

Low molecular weight heparin

The dose of Enoxaparin is based on most recent weight (need to check U&E's at booking appointment or DAU appointment, whichever is earliest):

Prophylactic dosing:

Current Weight	Creatinine clearance ≥ 30ml/min
< 50 kg	Enoxaparin 20 mg OD
50 – 89.9 kg	Enoxaparin 40 mg OD
90 – 129.9 kg	Enoxaparin 60 mg OD
130 – 169.9 kg	Enoxaparin 80 mg OD
≥ 170 kg	0.6 mg/kg/day

If creatinine clearance is less than 30ml / min – discuss with Consultant Haematologist via switchboard

Current Weight	Creatinine clearance < 30 ml / min
< 50 kg	Enoxaparin 20 mg OD*
50 – 89.9 kg	Enoxaparin 20 mg OD
90 – 129.9 kg	Enoxaparin 40 mg OD
130 – 169.9 kg	Enoxaparin 60 mg OD
≥ 170 kg	0.5 mg/kg/day

*monitoring of trough anti-Xa on day 5 -7 recommended. Target anti-Xa ≤ 0.1 units / ml.

Therapeutic Dosing:

For patients with CrCl ≥ 30ml/min:

Standard / OD dosing 1.5 mg / kg		High risk / BD dosing 1 mg / kg	
< 46 kg	60 mg OD	**< 44kg**	40 mg **BD**
47 - 59.9 kg	80 mg OD	**44 - 54.9 kg**	50 mg **BD***
60 - 73.9 kg	100 mg OD	**55 - 64.9 kg**	60 mg **BD**
74 - 89.9 kg	120 mg OD	**65 - 74.9 kg**	70 mg **BD***
90 - 109.9 kg	150 mg OD	**75 - 84.9 kg**	80 mg **BD**
110 - 125.9 kg	180 mg OD	**85 - 94.9 kg**	90 mg **BD***
126 - 139.9 kg	100 mg **BD**	**95 - 104.9 kg**	100mg **BD**
140 - 150 kg	112.5 mg **BD****	**105 - 114.9 kg**	112.5 mg **BD****
> 150 kg	Discuss Haematology***	**115 - 134.9 kg**	120 mg **BD**
		135 - 150 kg	150 mg **BD**
		> 150kg	Discuss Haematology***

* These doses require expulsion of 0.1 ml of enoxaparin from the next syringe size above using the gradations marked on the syringe. If this is not practical for outpatient dosing then discuss use of whole syringes or alternatives with Pharmacy / Haematology.

**These doses require expulsion of 0.05ml of the enoxaparin 120mg/0.8ml syringe using the gradations marked on the syringe.

***Dosing will generally be 0.75 mg / kg BD for standard indication and 1 mg / kg BD for high risk indications rounded to appropriate vial size.

For CrCl < 30 ml / min: discuss with Consultant Haematologist via switchboard

Weight	**CrCl** < 30 ml / min
< 45 kg	40 mg enoxaparin OD
45 - 54.9 kg	50 mg enoxaparin OD*
55 - 64.9 kg	60 mg enoxaparin OD
65 - 74.9 kg	70 mg enoxaparin OD*
75 - 84.9 kg	80 mg enoxaparin OD
85 - 94.9 kg	90 mg enoxaparin OD*
95 - 109.9 kg	100 mg enoxaparin OD
110 - 134.9 kg	120 mg enoxaparin OD
135 - 159.9 kg	150 mg enoxaparin OD
160 - 190 kg	180 mg enoxaparin OD

Monitor trough anti-Xa level after 5 - 7 days within 2 hours before the next dose, do not wait for the result before giving the dose. Target trough level is < 0.4 units / ml. If above range confirm timing of the injection and level and reduce by 1 dose level. Recheck level on new dose after 2 - 4 days.

*These doses require expulsion of 0.1 ml of enoxaparin from the next syringe size above using the gradations marked on the syringe. If this is not practical for outpatient dosing then discuss use of whole syringes or alternatives with Pharmacy / Haematology.

In women identified as having a bleeding risk, the balance of bleeding and clotting should be discussed with the Consultant Obstetrician and Consultant Haematologist.

Heparin induced osteopenia is a risk with prolonged use. Bone loss is similar to that experienced during a six-month period of breast feeding.

Venous thromboembolism prevention. Women should have a VTE risk assessment performed during pregnancy. COVID-19 is a transient risk during the 14 days around infection. Women who are self-isolating at home should stay hydrated and mobile. A risk assessment should be performed using the antenatal risk VTE risk assessment pathway – and COVID-19 scores +1. Women with a score ≥3 should be offered prophylactic LMWH for 14-21 days, which can be stopped thereafter (unless a persisting score ≥4 before 28 weeks or persisting score ≥3 after 28 weeks).

Management for delivery

- Women once in established labour should be advised not to inject any further heparin.
- If a woman on thromboprophylaxis is admitted for a planned delivery, she must be told to stop treatment 24 hours before admission
- Regional anaesthetics should not be given to a woman until
  - at least 12 hours after the last dose of prophylactic low molecular heparin
  - at least 24 hours after the last dose of therapeutic low molecular heparin
- Epidural catheters should not be removed within 12 hours of the most recent injection.
- Prophylactic doses of low molecular weight heparin should be given 3 hours after operative delivery (or 4 hours after regional anaesthetic techniques).

Postpartum prophylaxis

Risk management for all cases is mandatory.

Postnatal VTE assessment should be completed on EPR under maternity VTE risk assessment where separate proformas are available for antenatal and postnatal assessment. Appropriate advice given depending upon the risk (community staff will have access to EPR and they need to do the risk assessment if they see these women postnatally in the community).

This should happen as soon after delivery as possible and definitely within the first six hours. Each of the following risk factors is listed and should be marked with a tick if present. The number of ticks will help classify the patient into three different categories (low, intermediate and high risk).

Reassessment should occur if the clinical picture changes in the postnatal period, for example a mid-cavity delivery with no other risk factors has a pph on the ward of over 1 litre. To do this reassessment use the postnatal VTE risk assessment proforma on EPR.

Please note that;

- **Elective cases** should have Flowtrons as mechanical management until commencement of Enoxaparin or mobilisation (whichever comes first).
- **Emergency cases** that already have TEDs on should then go to Enoxaparin post op if required. **Emergency cases** with no TEDs on should have Flowtrons in theatre and then go on to Enoxaparin.
- Women with reasons for delayed administration of Enoxaparin (e.g. uterine tamponade) should have Flowtrons until Enoxaparin commences.

TED's and Flowtrons MUST NOT be used together – using the above criteria choose one or the other.

Risk assessment scoring system:

PERSONAL HISTORY	MEDICAL HISTORY
• Age >35 years • Parity ≥ 3 • BMI 30 - 40 • BMI >40 (= 2 risk factors) • Immobility e.g. Wheelchair bound • Smoking • Family history VTE (1st degree relative)	• Previous DVT/PE (Recurrent/ unprovoked/ oestrogen-related) • Previous DVT/PE (Provoked) • Thrombophilia Group 1 (Antithrombin deficiency; FVL homozygous; >1 thrombophilic defect; Antiphospholipid syndrome) • Thrombophilia Group 2 (FVL heterozygous; Protein C/S deficiency ;Prothrombin gene) • SLE/ Sickle cell disease • Nephrotic syndrome (proteinuria >3g/day) • Inflammatory bowel disease • Severe cardiac disease • Current malignancy • IV drug user • Gross varicose veins (symptomatic/ above knee/with phlebitis, oedema/skin changes)
CURRENT PREGNANCY	**DELIVERY**
• Dehydration • Severe PET • Multiple pregnancy or IVF • Severe systemic infection • Immobility • VTE in current pregnancy	• Prolonged labour (>24 hours) • Mid-cavity or rotational forceps • Legs in lithotomy >1hr • Caesarean section: elective or emergency • EUA/ERPC • 1º / 2º PPH (>1litre or blood transfusion)

Management plan:

Perform the risk assessment as per the risk assessment chart on EPR and plan whether women need thromboprophylaxis.

Timing of administration of thromboprophylaxis

Postpartum thrombophylaxis should be given as soon as possible after delivery but should be withheld for 4-6 hours after insertion or removal of epidural catheter.

Refer to the following tables for guidance on dosage according to patient's postnatal or most recent weight (weight at booking should guide if current weight not known)

Current Weight	Creatinine clearance ≥ 30 ml / min
< 50 kg	Enoxaparin 20 mg OD
50 – 89.9 kg	Enoxaparin 40 mg OD
90 – 129.9 kg	Enoxaparin 60 mg OD
130 – 169.9 kg	Enoxaparin 80 mg OD
≥ 170 kg	0.6 mg/kg/day

If creatinine clearance is less than 30ml / min – to discuss with Consultant Haematologist via switchboard

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130 – 169.9 kg	Enoxaparin 60 mg OD
≥ 170 kg	0.5 mg/kg/day

*Monitoring of trough anti-Xa on day 5 -7 recommended. Target anti-Xa ≤ 0.1 units / ml.

Patients should receive at least 6 weeks of postnatal thromboprophylaxis if they are assessed to be **high risk**.

Risk factors for **high risk**:

- Any previous VTE
- Anyone requiring antenatal LMWH
- High risk thrombophilia
    - Anti-thrombin deficiency
    - Protein C deficiency
    - Protein S deficiency
- Low risk thrombophilia +family history

Patients who are at **intermediate risk** should receive at least 10 days postnatal thromboprophylaxis. Consider extending thromboprophylaxis for those with persisting or >3 cumulative risk factors.

Risk factors for **intermediate risk**:

- Caesarean section in labour
- BMI>40kg/m^2
- Readmission or prolonged admission (≥ 3 days) in the puerperium
- Any surgical procedure in the puerperium except immediate repair of the perineum
- Medical co-morbidities:
    - Cancer
    - Heart failure
    - Active SLE
    - Inflammatory bowel disease or inflammatory polyarthropathy
    - Nephrotic syndrome
    - Type 1 DM with nephropathy
    - Sickle cell disease
    - Current IVDU

Patients with 2 or more cumulative risk factors should receive postnatal thromboprophylaxis as per **intermediate risk** (see above).

Patients with fewer than 2 cumulative risk factors are at **lower risk** and require early mobilisation and avoidance of dehydration.

Cumulative risk factors:

- Age > 35 years
- Obesity (BMI ≥ 30kg/m^2)
- Parity ≥ 3
- Smoker
- Elective caesarean section
- Family history of VTE
- Low-risk thrombophilia
- Gross varicose veins
- Current systemic infection
- Immobility, e.g. paraplegia, PGP, long-distance travel
- Current pre-eclampsia
- Multiple pregnancy
- Preterm delivery in this pregnancy (<37^{+0} weeks)
- Stillbirth in this pregnancy
- Mid-cavity rotational or operative delivery
- Prolonged labour (>24 hrs)
- PPH > 1 litre or blood transfusion

Prescriptions for the entire postnatal course of low molecular weight heparin should be issued in secondary care (MBRRACE 2023).

References:

1. MBRRACE 2023 report – Saving lives, Improving Mother's care. Lessons learned to inform maternity care from the UK and Ireland confidential inquiries to maternal deaths and morbidity 2019 -2021. https://www.npeu.ox.ac.uk/assets/downloads/mbrrace-uk/reports/maternal-report-2023/MBRRACE-UK_Maternal_Compiled_Report_2023.pdf
2. Royal College of Obstetricians and Gynaecologists. Reducing the Risk of Venous Thromboembolism during Pregnancy and the Puerperium GTG 37a (amended 2023). London: RCOG. Available at: https://www.rcog.org.uk/media/m4mbpjwi/gtg-no37a-2015_amended-2023.pdf
3. Royal College of Obstetricians and Gynaecologists. (2015). Thromboembolic Disease in Pregnancy the Puerperium: Acute Management London: RCOG. Available at: https://www.rcog.org.uk/media/wj2lpco5/gtg-37b-1.pdf
4. National Institute for Health and Clinical Excellence (NICE). (Jan 2024). Caesarean birth (NG192). guidance London: NICE. Available at: https://www.nice.org.uk/ /ng192
5. National Institute for Health and Clinical Excellence (NICE). (2014). Intrapartum Care: (NG235). London: NICE. Available at: https://www.nice.org.uk/guidance/ng235
6. Antenatal pulmonary embolism, Risk factors, management and outcomes. BJOJ: M Night, UKOSS.Volume115, Issue 4, March2008, page 453-461 https://obgyn.onlinelibrary.wiley.com/doi/abs/10.1111/j.1471-0528.2007.01622.x

Auditable standards:

1. All women will have a VTE risk assessment done at booking by fully completing the antenatal boxes in VTE risk assessment form or at first antenatal clinic visit. The appropriate actions will be carried out following the risk assessment.
2. All women admitted to maternity (inpatient) will have a VTE risk assessment done per admissions by fully completing the relevant boxes (antenatal / delivery) in VTE risk assessment on EPR. The appropriate actions will be carried out following the risk assessment.
3. All women who require thromboprophylaxis or treatment for a diagnosis of VTE will have an individual management plan documented in the maternal record on EPR.
4. All women who had thromboprophylaxis in the antenatal period will be advised to stop it 24 hours before admission for planned deliveries.